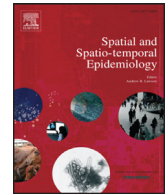




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Original Research

A note on intrinsic conditional autoregressive models for disconnected graphs



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ABSTRACT

In this note we discuss (Gaussian) intrinsic conditional autoregressive (CAR) models for disconnected graphs, with the aim of providing practical guidelines for how these models should be defined, scaled and implemented. We show how these suggestions can be implemented in two examples, on disease mapping.

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1. Introduction

Conditional Autoregressive (CAR) models are widely used to represent local dependency between random variables, with numerous applications in spatial statistics, disease mapping (Lawson, 2013; Wakefield, 2007) and imaging (Besag et al., 1991). This paper discusses CAR models for disconnected graphs and provides specific recommendations in order to implement them in practice. Through this note, we will illustrate the proposed methods using a classic example on disease mapping, leaving the straightforward generalisation to the reader.

Disease mapping concerns the study of disease risk over a map of geographical regions. Let assume a study area partitioned in n non overlapping regions, indexed by $i = 1, \dots, n$, with y_i the number of cases observed for a given disease in region i . If the disease is rare, y_i can be as-

sumed as Poisson distributed with mean $\theta_i = E_i r_i$, where E_i is the expected number of cases (computed on the basis of the demographic characteristics of a reference population) and r_i is the relative risk associated with living in region i ; $r_i = 1$ means no augmented risk with respect to the average in the whole study area, $r_i > 1$ ($r_i < 1$) indicates higher (lower) risk than average.

As an example, in ecological regression studies investigating the relationship between pollution, say z , and health, the risks might be modelled as

$$\log(r_i) = \alpha + \beta z_i + x_i$$

where α is the log baseline risk and β is the effect of pollution. The term x_i is a random effect capturing residual extra variability in i , possibly due to unobserved risk factors. Some of these unobserved may vary smoothly over space, inducing spatial structure in the residuals x_i 's. This structure can be modelled using a CAR prior.

The definition of a CAR model starts by specifying a graph. The graph consists of a collection of nodes and edges representing, respectively, regions and neighbouring

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relationships between them. The number of nodes determines the size of the graph. For the sake of a general definition, in this paper a graph is seen as a collection of nodes belonging to one or more *connected components*.

Within a connected component, each node is connected to at least another node through a path (i.e. a set of contiguous edges). Therefore, we say a graph is *connected* if it consists of one connected component of size larger than one (e.g. a ‘mainland’ component). A graph is *disconnected* if it is not connected, meaning that it consists of more than one connected component of any size (e.g. a ‘mainland’ component of size n plus an ‘island’ component of size 1).

Within a connected graph, specification of neighbouring relationships is clear, as each node has at least one neighbour. In this case, the definition of a CAR model follows straightforwardly from specification of the full conditionals (Besag et al., 1991)

$$x_i | \mathbf{x}_{-i}, \kappa \sim \mathcal{N} \left(\sum_{j: i \sim j} x_j / n_i, (n_i \kappa)^{-1} \right), \quad i = 1, \dots, n. \quad (1)$$

where $\mathbf{x}_{-i} = \{x_k, k \neq i\}$, $i \sim j$ means i and j are neighbouring nodes and n_i is the number of neighbours of i . The precision parameter κ regulates the degree to which x_i is shrunk to the local mean $\sum_{j: i \sim j} x_j / n_i$. Note that the variance $\text{Var}(x_i | \mathbf{x}_{-i})$ is inversely proportional to the number of neighbours n_i . Therefore, both the prior assigned to κ and the structure of the graph play a role in determining the shrinkage properties of the CAR prior.

The graph structure is essentially defined through conditional independence assumptions between the nodes. In particular, prior (1) specifies that the x_i 's are conditionally independent given the information in neighbourhood x_j , $i \sim j$. These assumptions are reflected in the precision structure of the joint distribution for $\mathbf{x} = (x_1, \dots, x_n)^T$, derived from the full conditionals in (1),

$$\pi(\mathbf{x} | \kappa) = \left(\frac{\kappa}{2\pi} \right)^{(n-1)/2} |\mathbf{R}|_*^{1/2} \exp \left(-\frac{\kappa}{2} \sum_{i \sim j} (x_i - x_j)^2 \right), \quad (2)$$

where the summation is over the set of all pairs of neighbours, $i \sim j$, and $|\cdot|_*$ represents the generalised determinant, calculated as the product of the non zero eigenvalues. Model (2) is a multivariate Gaussian with zero mean and $n \times n$ precision $\mathbf{Q} = \kappa \mathbf{R}$, where $\mathbf{R}$ is a matrix representing the neighbourhood structure of the model:

$$R_{ij} = \begin{cases} n_i & i = j \\ -1 & i \sim j \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

The conditional independence assumptions between the nodes can be checked immediately looking inside $\mathbf{R}$: if $R_{ij} = 0$ then x_i and x_j are conditionally independent, given all the other variables $\{x_k: k \neq i, k \neq j\}$ (Rue and Held, 2005). The structure matrix in (3) has rank equal to $n - 1$, thus density (2) is improper. As a consequence of this, the overall mean is unspecified in Eq. (2) and can be identified only when adding a linear constraint, such as $\sum_i x_i = 0$. For this

reason, this type of prior is referred to as intrinsic CAR. The purpose of using an intrinsic CAR prior is to borrow strength of information between neighbours, in the disease mapping example this yields a smoothed map for the risk.

In some applications the graph might be disconnected. A typical example is a graph made of two connected components, one of size larger than 1 (e.g. mainland) and the other of size 1 (e.g. island; components of size 1 will be denoted as singletons hereafter). In this case, a direct application of the intrinsic CAR (2) implies that $n_i = 0$, if i is a singleton. This yields a constant prior for x_i , i.e. the singleton random effect is, at prior, allowed infinite variance. This poses a general issue about interpretation of the prior for κ , that may be different in the various connected components of the graph. These issues will be discussed in details in Section 3, where we will introduce the general case of a disconnected graph, with several connected components of any size.

Motivated by the issues involved in a direct application of (2) to a disconnected graph, we propose a solution based on rescaling the precision matrix $\mathbf{Q}$ to have similar shrinkage properties, at prior, in each connected component. The rescaling procedure is drawn by ideas in Sørbye and Rue (2014). With our new definition the effect of the graph on the shrinkage properties (defined by the prior for κ) is marginalized out. Therefore, κ has a clear interpretation as a smoothing parameter, regulating the degree to which x_i will shrink to a local mean, if i has neighbours, and to a global mean, if i has no neighbour. In literature there is a lack of attention (Knorr-Held, 2002) on the definition and/or proprieties of a CAR for disconnected graphs, the only reference on this topic is Hodges et al. (2003) who discuss the form of the normalizing constant in (2). On the practical side, the GeoBUGS manual (Spiegelhalter et al., 2002) offers some guidelines in the case of a graph containing singletons, with a default option which is to set x_i to zero, if i is a singleton. Note that this is equivalent to enforcing a sum-to-zero constraint ($x_i = 0$) on each singleton random effect.

An alternative approach is to *correct* the graph, i.e. to remove the singletons by connecting islands to mainland. In our opinion neither of these strategies address the issue in a satisfactory way: the first one adds new constraints, the second one essentially corrupts the graph. We would like to stress that the definition of the graph is part of the modelling process, therefore editing new edges between islands and mainland is only appropriate in cases where borrowing strength of information between them is needed. Changing the graph is inappropriate every time the application at hand requires the original graph, i.e. when borrowing strength of information between islands and mainland is not a sensible choice.

The rest of the paper is organized as follows. Section 2 introduces formally the concept of an intrinsic CAR model defined with respect to a graph. In Section 3, we revise scaling of the precision matrix of an intrinsic CAR model defined with respect to a connected graph, following Sørbye and Rue (2014). In Section 4 we discuss the issues caused by direct application of (2) in the case of a disconnected graph. We then outline recommendations on how the model should be scaled in this case.

Section 5 deals with linear constraints and computation of the normalizing constant. In Section 6 we illustrate the proposed methods in two examples on disease mapping involving two different types of graphs. We conclude with a discussion in Section 7.

2. Intrinsic CAR model defined with respect to a graph

In its simple form, the density of an intrinsic CAR model for $\mathbf{x}$ is

$$\pi(\mathbf{x} | \kappa) \propto \frac{1}{Z_n(\kappa)} \exp \left(-\frac{\kappa}{2} \sum_{i \sim j} w_{ij} (x_i - x_j)^2 \right) \quad (4)$$

where Z_n is a normalizing constant that we will return to later on and w_{ij} is a positive and symmetric weight assigned to the pair of neighbouring nodes i and j . The definition of a “neighbour” is application dependent and part of the model specification. For example, in many spatial applications, two regions (i and j , say) are considered to be neighbours if they share a common border. The weight w_{ij} might be chosen as the inverse Euclidean distance between the centroids of regions i and j . With no loss of generality, in this work we only discuss the case of unitary weights, $w_{ij} = 1$, if i and j share a common border, corresponding to Eq. (2).

The interpretation of distribution (4) is that similarity between two neighbours are encouraged, and this induces a smoothing effect between neighbours, and thereby between neighbours of neighbours, and so on. We can formalize this, by defining an undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, with a set of vertices $\mathcal{V} = \{1, 2, \dots, n\}$, and edges $\mathcal{E}$, the (unordered) set of neighbours. We then say that the intrinsic CAR is defined with respect to the graph $\mathcal{G}$.

The precision parameter κ determines the amount of smoothing and it is commonly estimated from data. The density (4) is improper or *intrinsic*, in the sense that it is invariant to adding the same constant to all the x_i 's, as an example of a first order polynomial intrinsic CAR. Higher order polynomial (and non-polynomial) intrinsic CAR's can be defined similarly; see (Rue and Held, 2005, Ch. 3) for a thorough discussion.

We will focus on discussing the simplest cases of intrinsic CAR models for graphs, displayed in Eq. (4), that are easily generalized to other types of intrinsic CAR models. This approach avoids technicalities which may distract from understanding the underlying ideas.

3. Scaling of an intrinsic CAR defined for a connected graph

Intrinsic CAR models have an unresolved issue with *scaling*, which is not addressed by studying (4). The basic reference is Sørbye and Rue (2014), on which we will base our arguments in this section.

Let us start by assuming that the graph $\mathcal{G}$ is connected, meaning that it is made of one connected component. An example of a connected graph is shown in Fig. 1a. The intrinsic CAR model (4) defined for this graph, has precision

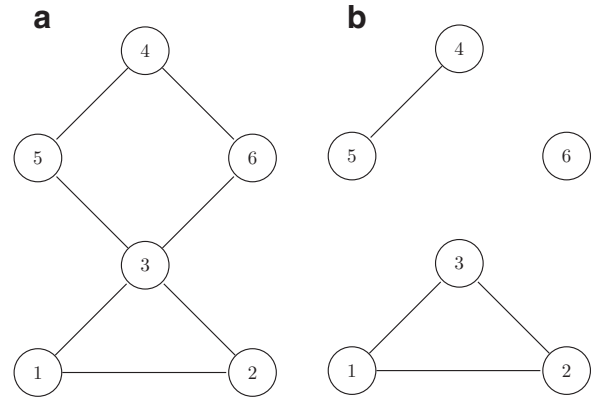


Fig. 1. The two graphs used in the discussion: (a) a connected graph, and (b) a disconnected graph.

matrix (the zeros are not shown):

$$\mathbf{Q} = \kappa \begin{bmatrix} 2 & -1 & -1 & & & \\ -1 & 2 & -1 & & & \\ -1 & -1 & 4 & & & \\ & & & 2 & -1 & -1 \\ & & & -1 & 2 & \\ & & & -1 & -1 & 2 \end{bmatrix}. \quad (5)$$

The matrix is singular with rank deficiency of 1. The intrinsic density is invariant to adding a constant to $\mathbf{x}$, meaning that $\mathbf{x}$ and $\mathbf{x} + c\mathbf{1}$ has the same improper density. However, what is of practical interest here, is *how* and by *how much* this model varies around its mean value $\bar{\mathbf{x}}$, i.e. how $\mathbf{x}$ varies if we impose a sum-to-zero constraint $\mathbf{1}^T \mathbf{x} = 0$. The critical issue is that this is a complicated function of the graph $\mathcal{G}$, for which we have no good intuition.

Let us formalize this by introducing the concept of *typical marginal variance*. Let i be a node of a connected graph of size n , we denote as $s_i^{(k)}$ the (conditional on κ) marginal variance of x_i , which is the i^{th} entry in the diagonal of the generalised inverse of $\mathbf{Q}$. The typical marginal variance of an intrinsic CAR defined with respect to a connected graph is defined as the geometric mean of the marginal variances,

$$S^{(\kappa)} = \exp \left(\frac{1}{n} \sum_{i=1}^n \log(s_i^{(\kappa)}) \right)$$

We assume $S^{(\kappa)}$ as a measure of the deviation of $\mathbf{x}$ from its null space, i.e. after applying the constraint $\mathbf{1}^T \mathbf{x} = 0$. Note that this quantity depends on both κ and the graph, through the structure $\mathbf{R}$.

For our example the marginal variances, s_i 's, are $0.53/\kappa$, $0.53/\kappa$, $0.19/\kappa$, $0.53/\kappa$, $0.44/\kappa$ and $0.44/\kappa$ for $x_1, \dots, x_6$, which we interpret as follows.

- The issue - that marginal variances are different - is a *feature* of the intrinsic CAR model, consequently of that the conditional variance, $\text{Var}(x_i | \mathbf{x}_{-i})$, is inversely proportional to the number of neighbours of node i , see Eq. (1).
- The typical marginal variance is confounded with our interpretation of the precision parameter κ , and we do

not know a-priori what $\kappa = 1$ means in terms of a typical marginal variance. In the Bayesian framework, this is a crucial issue, since we need to impose a prior distribution for κ . We need to address what κ means in terms of its impact on the model.

The solution proposed in Sørbye and Rue (2014) of this apparent dilemma, is simply to scale $\mathbf{Q}$ so that the typical marginal variance is 1 when $\kappa = 1$. They also recommend multiplying $\mathbf{Q}$ by the typical marginal variance calculated for $\kappa = 1$, i.e. $S^{(1)} = \exp\left(\frac{1}{n} \sum_{i=1}^n \log(s_i^{(1)})\right)$. In our example, applying this scaling yields the following scaled precision matrix,

$$\mathbf{Q}_{\text{scaled}} = \kappa S^{(1)} \begin{bmatrix} 2 & -1 & -1 & & & \\ -1 & 2 & -1 & & & \\ -1 & -1 & 4 & & & \\ & & & 2 & -1 & -1 \\ & & & -1 & -1 & 2 \\ & & & -1 & -1 & 2 \end{bmatrix} \quad (6)$$

where $S^{(1)} = 0.4219$. The most important consequence of this scaling, is that κ is now the typical precision and not only a precision parameter. This makes it possible to define a meaningful prior distribution and have clear interpretation for κ . There is a long tradition to prefer model parameters with a good and clear interpretation.

Our recommendation is clear and unambiguous.

Recommendation 1. We recommend scaling intrinsic CAR models defined with respect to connected graphs.

Computation of the scaling quantity $S^{(1)}$ needs the generalised inverse of $\mathbf{R}$ (recall $\mathbf{Q}$ is equal to $\mathbf{R}$, if $\kappa = 1$), which is an $\mathcal{O}(n^3)$ operation. A better approach, is to make use of the graph of the model and the knowledge of the null-space, and treat $\mathbf{R}$ as a sparse matrix; see Rue and Held (2005, Ch. 2.4) for background and details. The scaling $S^{(1)}$ can then be computed from a rank one correction of the marginal variances from the unconstrained model, see Rue and Martino (2007) for technical details about the recursions leading to the marginal variances. The computational cost will be $\mathcal{O}(n^{3/2})$ for typical spatial graphs, which is a huge improvement. The R-function `inla.scale.model()` in the R-INLA package (see www.r-inla.org) is an efficient implementation of this.

4. Scaling of an intrinsic CAR defined for a disconnected graph

A graph is disconnected if it is not connected, i.e. it is made of more than one connected component of any size. The practical interpretation of this related to (4), is that there are separated regions and/or nodes with no neighbours. Such cases easily appear if the graph is constructed from a map where we can have islands, or regions that are physically disconnected from the rest of the area. Fig. 1b shows a disconnected graph with three *connected components* of size 3, 2 and 1 (i.e. a singleton). We will use this graph as reference, in this section.

A direct application of the intrinsic CAR model for graph in Fig. 1b, gives the precision matrix

$$\mathbf{Q} = \kappa \begin{bmatrix} 2 & -1 & -1 & & & \\ -1 & 2 & -1 & & & \\ -1 & -1 & 2 & & & \\ & & & 1 & -1 & \\ & & & -1 & 1 & \\ & & & & & 0 \end{bmatrix}. \quad (7)$$

This matrix is singular with rank-deficiency of 3, since the density is invariant when we add a constant to each connected component. There are several unfortunate issues with this intrinsic CAR model, simply because the *implicit* assumption behind (4) is that the graph is connected. Similar to Section 3, we have the following issues due to the disconnected graph.

- Node 6 has no neighbours so $Q_{6,6} = 0$ and we have a constant density for x_6 . This can lead to an improper posterior distribution. To see this, consider the following model for an observed count, y_6 , in node 6,

$$y_6 | x_6 \sim \text{Po}(\exp(x_6))$$

$$\pi(x_6 | \kappa) \propto \text{const} \quad (8)$$

where $x_6 = \log(\theta_6)$ is the Poisson mean, θ_6 , in logarithmic scale. The constant prior (8) implies an improper prior on the Poisson mean, i.e. $\pi(\theta_6) \propto 1/\theta_6$. If a zero count is the case, the posterior $\pi(\theta_6 | y_6 = 0) \propto \exp(-\theta_6)/\theta_6$ is an improper density. In other words, the constant prior for the singleton makes it difficult for the singleton random effect to shrink to the global mean. Also, this goes against the purpose of using (4) in the first place, which is to do smoothing and borrowing strength.

- The connected components for (x_1, x_2, x_3) and (x_4, x_5) are defined as on a connected graph. Even though this is reasonable within each connected component, it is not reasonable when we compare across them. The issue is that the typical marginal variance will in general be different between the connected component. Following Section 3, the marginal variances are $0.22/\kappa$, $0.22/\kappa$, $0.22/\kappa$, $0.25/\kappa$, $0.25/\kappa$ and ∞ for $x_1, \dots, x_6$. (Note that in this simple case the marginal variances are constant within each connected component.) The typical marginal variance for the first and second component is $S_1^{(1)} = 0.22$ and $S_2^{(1)} = 0.25$, while it is ∞ for the singleton x_6 .

To resolve both these issues we propose scaling the intrinsic CAR with respect to the disconnected graph, i.e. scaling (7) similarly to Section 3, but with two minor modifications.

1. We scale each connected component of size larger than one, independently as described in Section 3, (i.e. we apply recommendation 1 to each connected component of size larger than one).
2. For connected components of size one, we replace the constant prior with a standard Gaussian with precision κ . This assumes the singleton as a non spatial random effect with precision κ , like in Wakefield (2007).

In our example, application of the two instructions above produce the following scaled precision matrix:

$$Q_{\text{scaled}} = \kappa \left(S_1^{(1)} \begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix} \begin{matrix} 0 & 0 \\ 0 & 0 \\ 0 \end{matrix} \right) + S_2^{(1)} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} 1 & -1 \\ -1 & 1 \\ 0 \end{matrix} + \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} 0 & 0 \\ 0 & 0 \\ 1 \end{matrix} \right)$$

where $S_1^{(1)}$ and $S_2^{(1)}$ are the scaling values for, respectively, the first and second connected component of size larger than one. Note that, since a standard Gaussian prior with precision κ is assigned to the singleton, its marginal variance is the same as for two the connected components. This scaling gives a well defined interpretation of the precision parameter κ . The prior for κ has (approximately) the same impact on the different connected components of the graph.

Our recommendation is clear and unambiguous.

Recommendation 2. We recommend scaling intrinsic CAR models defined with respect to disconnected graphs.

The popular WinBUGS software treats singletons as non-stochastic nodes and sets the associated random effects to zero by default (see GeoBUGS manual (Spiegelhalter et al., 2002, page 18)). From the perspective of developing user friendly software to fit Bayesian models via Markov Chain Monte Carlo (MCMC) algorithms this seems a safe strategy: if singletons are taken as stochastic nodes, with consequent improper $\pi(x_i) \propto \text{const}$, this may lead to poor mixing and extremely slow convergence especially when a zero count is observed.

5. Linear constraints and normalising constants

When using intrinsic models we always have to be careful not to introduce confounding (Goicoa et al., 2018). Let $\mathbf{x}$ be the intrinsic CAR defined on a connected graph, then the linear predictor

$$\eta = \mathbf{x}$$

and

$$\eta = \mu \mathbf{1} + \mathbf{x} | (\mathbf{1}^T \mathbf{x} = 0)$$

are the same. In the first case, there is no intercept as it is implicit in the null-space of the precision matrix for $\mathbf{x}$.

In the second case, we explicitly define the intercept and remove it from the intrinsic CAR model. We strongly prefer the second option, since it makes the interpretation of each component explicit and reduces the chance of misunderstanding and misspecifying the intrinsic CAR in more complex scenarios than here.

When the graph is disconnected, we designate one intercept for each connected component with size larger than one. If a connected component specific intercept is needed, we prefer to add it to the model explicitly rather than implicitly.

Recommendation 3. We recommend to use a sum-to-zero constraint for each connected component of size larger than one.

Rue and Held (2005, Ch 3) provide a strong case, to interpret the normalizing constant for the proper part of the model and the improper part as a diffuse Gaussian. Assume the graph has n_c connected components, each of size n_i . Then the normalizing constant for the scaled intrinsic CAR model (4), will be

$$Z_n(\kappa) = |\mathbf{R}|_*^{1/2} \prod_{i=1}^{n_c} Z_{n_i}(\kappa)$$

where

$$Z_n(\kappa) = \begin{cases} (\kappa/(2\pi))^{1/2} & \text{if } m = 1 \\ (\kappa/(2\pi))^{(n-1)/2} & \text{otherwise.} \end{cases}$$

and $|\cdot|_*$ is a generalised determinant defined as the product of all non-zero eigenvalues. In most cases, we only need the part of $Z_n(\kappa)$ that depends on κ , hence we do not need to compute the generalised determinant nor carry the 2π around.

6. Application

In this section we provide two applications, first we show how the scaling works with a disconnected graph, using lip cancer data from Scotland when we have three islands. In the second application, we show how to include specific component intercepts, using stomach cancer data from Tuscany (Italy) where we have two islands and two connected components.

6.1. Scottish Lip Cancer data: a graph with three singletons

The data are counts of lip cancer cases registered in $n = 56$ Scottish counties during years 1975–1980. Fig. 2a shows the standardized mortality ratios (SMRs) over the regions. The graph is displayed in Fig. 2b, it has been constructed assuming the counties are the nodes, with edges connecting counties sharing borders. Three counties are island (Orkneys, Shetland and the Outer Hebrides). We apply the model used in Clayton and Breslow (1993). Let assume vectors y_i and E_i are, respectively, observed and expected lip cancer cases during the study period in regions i , covariate z is the “percentage of the population engaged in agriculture, fishing, or forestry” (AFF) and r_i the unknown relative risks, the model is

$$y_i \sim \text{Po}(E_i r_i) \quad (9)$$

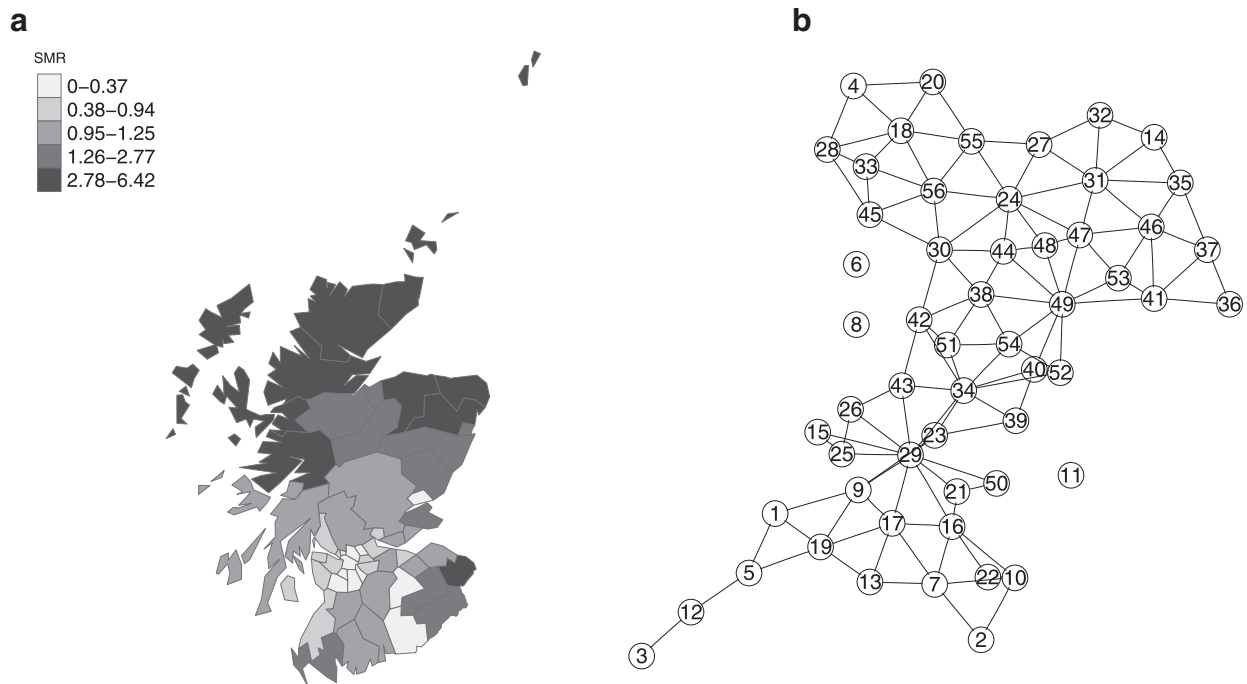


Fig. 2. Scotland map (a) and (b) the disconnected graph induced by the map.

$$\log(r_i) = \alpha + \beta z_i/10 + x_i \quad (10)$$

$$\pi(\mathbf{x} | \kappa) \propto \frac{1}{Z_n(\kappa)} \exp\left(-\frac{\kappa}{2} \mathbf{x}^T \mathbf{R} \mathbf{x}\right). \quad (11)$$

Clayton and Breslow (1993) analysed the spatial dependency using an intrinsic CAR model defined on a connected graph, obtained by editing new edges to connect islands to mainland. Following the natural discontinuity between islands and mainland, we illustrate our definition of an intrinsic CAR with respect to the disconnected graph in Fig. 2b. This graph is made of four connected components, three of them are singletons.

Our aim is not to compare models built on different graphs, to evaluate which one fits best. In this first example, our aim is to compare results obtained under the scaled intrinsic CAR model and the unscaled one and emphasize the benefit of scaling. Recall the scaled model is obtained by scaling the precision $\mathbf{Q}$ (recommendations 2) as described in detail in Section 4, while the unscaled model has precision $\mathbf{Q} = \kappa \mathbf{R}$, where $\mathbf{R}$ is defined as in (3), i.e. $\mathbf{R}[i, i] = 0$ if i is an island. For the sake of comparison, for both scaled and unscaled models, we use the same prior $\pi(\kappa)$ (gamma with shape 1 and rate $5e-5$) and apply the sum-to-zero constraint $\mathbf{1}^T \mathbf{x} = 0$ to the nodes of the connected component.

Fig. 3 displays the marginal posterior for six random effects fitted under the scaled (solid line) and unscaled (dashed line) model, using R-INLA. The scaled and unscaled models have different shrinkage properties, this is reflected in the large deviations in the top panels of Fig. 3. Note that the island-specific random effects estimated by

the unscaled model are less shrunk towards no effect, $x_i = 0$, than those estimated by the scaled model. On the other hand, the posterior for the three random effects belonging to the connected component of the graph (bottom panels of Fig. 3) are essentially the same. Results for the other nodes in the connected component are similar and not shown here. The different shrinkage properties of the two models are confirmed by looking at the posterior summaries in Table 1. Though the fixed effects α and β are almost unchanged, the relative risks for the singletons are more extreme under the unscaled model than the scaled one.

In the scaled model, the hyper-parameter κ has a clear interpretation as a typical precision. The marginal variance within each of the four components of the Scotland graph (the mainland and the three singletons) is κ^{-1} . In the unscaled model, the marginal variances in the different components are different, in particular it is infinite in the three singletons. For this reason we believe the more extreme estimates obtained under the unscaled model are due to the prior, not a feature of the data (i.e. overfitting). In general, how severe the overfitting will depend on the structure of the graph, the sample size and the prior assigned to κ . For instance, if data contain little information about the disease risk (e.g. low expected counts, which is not the case with the lip cancer data), the prior $\pi(\kappa)$ may have a large impact on the analysis and larger deviations between scaled and unscaled must be expected.

In Table 1, we reported also the Deviance Information Criterion (DIC), the number of effective parameters (pD) and the mean deviance $\bar{D}$. All are very close for the two scaled and unscaled models on the disconnected graphs, and they are not much different from those obtained for

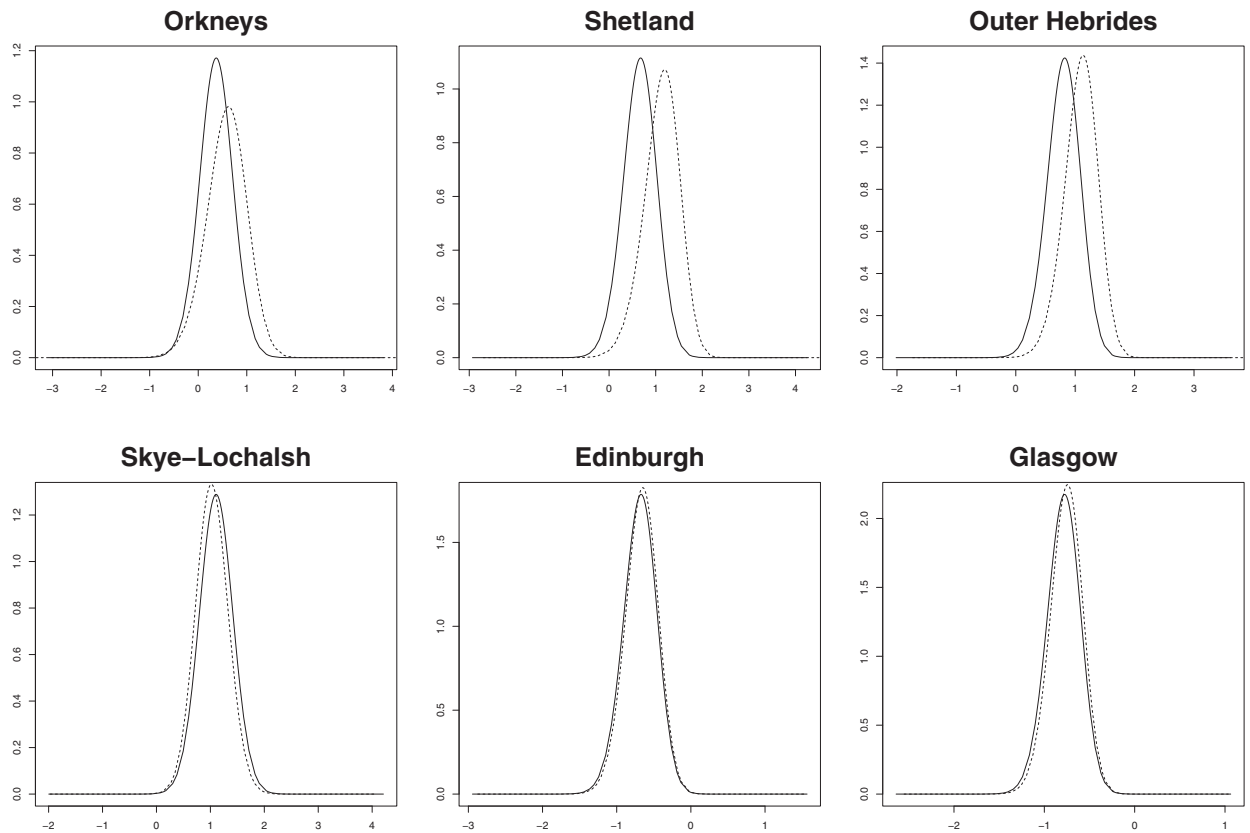


Fig. 3. The effect of scaling the disconnected graph. Upper panels show the marginal posterior (in the linear predictor scale) for the three singletons random effects, x_6 (Orkneys), x_8 (Shetland) and x_{11} (Outer Hebrides); lower panels show the marginal posterior for three nodes in the connected component of the graph, x_1 (Skye-Lochalsh), x_{45} (Edinburgh) and x_{49} (Glasgow). The marginals from the unscaled model (dashed lines) are less shrunk towards $x_i = 0$ than the marginals from the scaled model (solid line).

Table 1

Posterior summaries for scaled (top) and unscaled (bottom) versions of model (9)–(11) and Deviance Information Criterion (DIC), number of effective parameters pD and mean deviance $\bar{D}$.

Parameter	Mean	Standard deviation	2.5%	Median	97.5%
<i>Scaled model</i>					
κ (Precision)	3.97	1.17	2.16	3.81	6.69
α (Intercept)	−0.25	0.13	−0.50	−0.25	−0.00
β (AFF)	0.37	0.13	0.09	0.37	0.62
r_6 (Orkneys)	2.87	0.90	1.42	2.77	4.93
r_8 (Shetland)	2.06	0.73	0.95	1.95	3.80
r_1 (Outer Hebrides)	2.32	0.63	1.28	2.26	3.75
DIC	299.4				
pD	30.65				
$\bar{D}$	268.75				
<i>Unscaled model</i>					
κ (Precision)	2.26	0.70	1.18	2.15	3.91
α (Intercept)	−0.26	0.12	−0.50	−0.27	−0.02
β (AFF)	0.36	0.13	0.09	0.37	0.62
r_6 (Orkneys)	3.54	1.20	1.58	3.40	6.27
r_8 (Shetland)	3.26	1.18	1.36	3.11	5.96
r_1 (Outer Hebrides)	3.07	0.83	1.66	2.99	4.89
DIC	298.5				
pD	30.5				
$\bar{D}$	267.98				

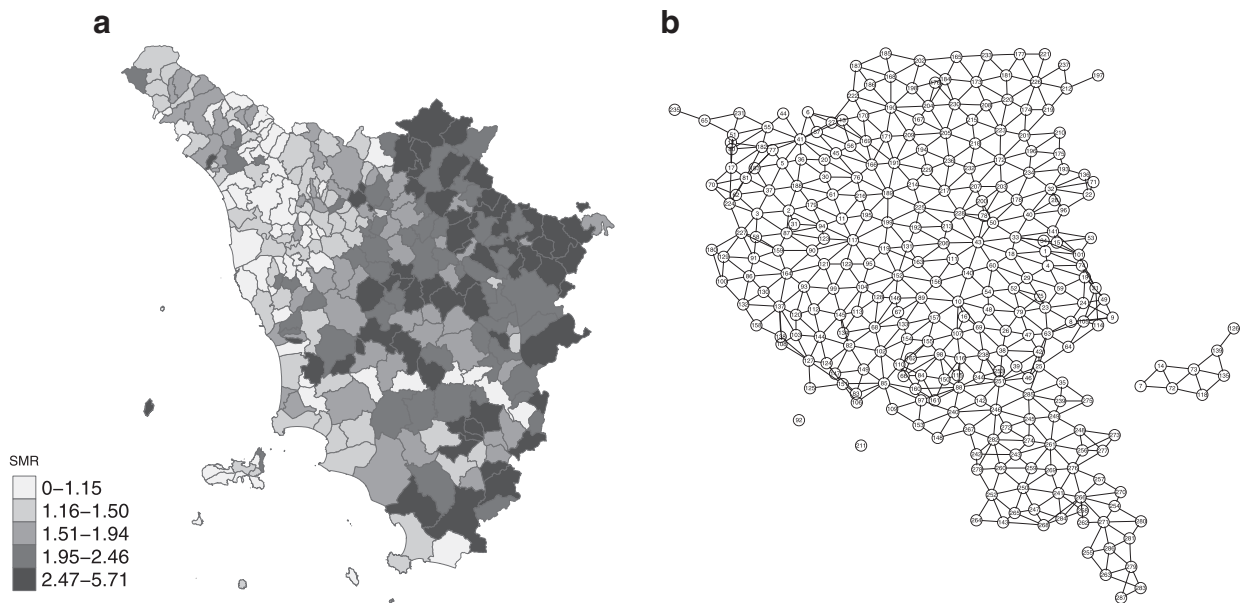


Fig. 4. Map of Tuscany (a) and the disconnected graph induced by the map (b).

Table 2

Statistical summaries: stomach cancer mortality for males over 35 years old, in Tuscany (Italy). Observed and expected, standardised mortality ratio (SMR) by census municipality 1971–1979.

		Observed	Expected	SMR
All($n = 287$)	5th percentile	2.3	1.93	0.69
	Median	18	9.30	1.68
	95th percentile	107	65.31	3.42
Elba($n = 8$)	5th percentile	1.35	2.87	0.34
	Median	4.00	4.9	0.82
	95th percentile	18.80	14.81	1.82
Without Elba ($n=279$)	5th percentile	3	1.90	0.72
	Median	18	9.50	1.71
	95th percentile	108	66.46	3.42

the scaled model on the connected graph: DIC was 297.1 ($pD = 28.7$ and $\bar{D} = 168.$). To allow this comparison we kept the same hyperpriors for all the models fitted.

6.2. Tuscany Lung cancer mortality: a graph split in sub-graphs

In this second example, we used data for stomach cancer deaths (9947 cases) in men over 35 years old, collected for each municipality between 1971–1979, from the Tuscan Cancer Atlas. Tuscan geography presents two small islands, that have separate municipalities, which we consider as singletons - Capraia and Giglio Isles - and one bigger island Elba composed of 8 municipalities and Tuscan mainland composed of 279 municipalities. Hence, the graphs is composed by 4 connected components, two of which are singletons (Fig. 4b). In Fig. 4a, we plotted the standardised mortality ratio (SMR) and observed that the highest SMRs are concentrated in the North-East.

In Table 2, we report statistical summaries for all the study regions, for Elba alone and for all mainland Tus-

can municipalities except Elba. In Elba there are 54 cases, hence lower values for observed, expected and SMR values.

The first model we fit is a Besag York Mollié (BYM) Besag et al. (1991), disease mapping model with scaled precision matrix on the disconnected graph, as follow:

$$y_i \sim \text{Po}(E_i r_i)$$

$$\log(r_i) = \alpha + x_i + v_i$$

Where $\mathbf{x}$ is modelled as a scaled intrinsic CAR with κ representing the precision as in Eq. (11) and $\mathbf{v}$ is the unstructured spatial residual $v_i \sim \mathcal{N}(0, \tau_v^{-1})$, with τ_v representing the precision of the unstructured random effects. Both precisions have as prior a gamma with shape 1 and rate $5e-5$.

In the second model, we include two separate intercepts one for the island of Elba and one for the Tuscan mainland (including the two singletons), to distinguish the lower SMRs observed in Elba versus the higher SMRs the rest of the region.

$$\log(r_i) = \alpha_j + x_i + v_i$$

Table 3

Posterior summaries for scaled one intercept (top) and scaled two intercepts, and Deviance Information Criterion (DIC), number of effective parameters pD and mean deviance $\bar{D}$.

Parameter	Mean	Standard deviation	2.5%	Median	97.5%
<i>Scaled Model-one intercept</i>					
τ_v (Heterogeneity)	10.350	2.150	6.399	10.296	14.767
κ (Precision)	122.574	224.899	15.536	61.987	606.1557
α (Intercept)	0.547	0.024	0.499	0.547	0.594
DIC	1818.21				
pD	182.24				
$\bar{D}$	1636.01				
<i>Scaled Model-Two intercepts</i>					
τ_v (Heterogeneity)	10.264	2.167	6.1929	10.253	14.576
κ (Precision)	157.446	343.495	17.07491	71.17663	832.5374
α_1 (Intercept Elba)	−0.048	0.193	−0.435	−0.045	0.324
α_2 (Intercept Tuscany)	0.558	0.025	0.510	0.558	0.606
DIC	1813.21				
pD	180.96				
$\bar{D}$	1632.22				

Where $j = (1, 2)$ indicating the two main connected components of size > 1 .

In Table 3, we report posterior model estimate summaries, with DIC, number of effective parameters and mean deviance. The results show that we should prefer the second model - the two intercepts- as the DIC is lower, and the intercept for Elba is 0.95 (CI 95% 0.64,1.38) while for the rest of the region the baseline risk is 1.74 (CI 95% 1.66,1.83). The unstructured effects remain unchanged between the two models, while precision κ , for the second model is higher, as assigning two intercepts help to explain better the spatial variability in the data. We compared these results with scaled and unscaled modes on a connected graph (scaled DIC = 1816.39 (pD = 180.7 and $\bar{D}$ = 1635.6 and unscaled DIC = 1816.63 (pD = 181.0 and $\bar{D}$ = 1635.6). For this data, the model with a separated intercept for each connected component was the best according DIC.

7. Summary

We motivated the definition of intrinsic CAR models for disconnected graphs under two main recommendations: scaling the precision structure and applying sum-to-zero constraints on the connected components of the graph. Scaling the precision structure sets the typical marginal variance to κ^{-1} in each component of the graph, where κ is the precision of the intrinsic CAR model. This immediately suggests a fair prior for random effects associated with the singletons in a disconnected graph in terms of a normal with zero mean and variance κ^{-1} . The advantage is that the prior assigned to κ has the same interpretation regardless of the particular structure of the graph.

We applied this strategy to a disease mapping example on lip cancer in Scotland, using the natural disconnected graph with three singletons. In this example we observed for the scaled model compared to the unscaled one, the random effect for the three islands were closer to the average mean.

In general, the extent to which the unscaled intrinsic CAR leads to overfitting should depend on the on the prior assigned on κ . When data contain little information about the disease risk for people living in the area (e.g. small regions with low expected counts, which is not the case with the lip cancer data), the prior $\pi(\kappa)$ may have a large impact on the analysis. In these situations larger deviations between scaled and unscaled intrinsic CAR models must be expected, as the unscaled one has no control on the marginal variance, hence no control on the impact of $\pi(\kappa)$, whereas the scaled one provides good intuition of κ .

Finally, in the second example, we show, how to distinguish the effect of each connected component by introducing separate intercepts for graph of size > 1 . The effect of jointly scaling the precision matrix and allowing specific separate baseline log risks for each component leaves the precision prior specification unchanged, as the scaling will guarantee that the connected components have the same precision assigned that regulates the degree to which the x_i are shrunk toward the local mean. When there is a belief that the component of the graph should not be linked jointly, the introduction of multiple intercept offers an elegant solution. An immediate application of this approach could be beneficial in epidemiological studies where geographical discontinuity is an issue.

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Update

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Erratum regarding missing Declaration of Competing Interest statements in previously published articles



Declaration of Competing Interest statements were not included in the published version of the following articles that appeared in previous issues of Spatial and Spatio-temporal Epidemiology

The appropriate Declaration/Competing Interest statements, provided by the Authors, are included below.

1. A tutorial on spatio-temporal disease risk modelling in R using Markov chain Monte Carlo simulation and the CARBayesST package (Spatial and Spatio-temporal Epidemiology, 2020 34C <https://doi.org/10.1016/j.sste.2020.100353> The authors were contacted after publication to request a Declaration of Interest statement.
2. A Bayesian latent process spatiotemporal regression model for areal count data (Spatial and Spatio-temporal Epidemiology, 2018 25-37 <https://doi.org/10.1016/j.sste.2018.01.003> The authors were contacted after publication to request a Declaration of Interest statement.
3. A note on intrinsic conditional autoregressive models for disconnected graphs (Spatial and Spatio-temporal Epidemiology, 2018 26C 34 <https://doi.org/10.1016/j.sste.2018.04.002> The authors were contacted after publication to request a Declaration of Interest statement.
4. Potential effects of climate change on the risk of accidents with poisonous species of the genus Tityus (Scorpiones, Buthidae) in Argentina (Spatial and Spatio-temporal Epidemiology, 2018; 25C 67-72) <https://doi.org/10.1016/j.sste.2018.03.002> The authors were contacted after publication to request a Declaration of Interest statement.
5. Geographical accessibility of community social services and incidence of self-harm (Spatial and Spatio-temporal Epidemiology, 2020;33C) <https://doi.org/10.1016/j.sste.2020.100334> The authors were contacted after publication to request a Declaration of Interest statement.

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